

VIETNAM NATIONAL UNIVERSITY HO CHI MINH CITY

HO CHI MINH CITY UNIVERSITY OF TECHNOLOGY

Faculty of Electrical and Electronics Engineering



Microwave Engineering and Systems

EE5430

Assignment 1

Transmission-Line Parameters and Wave Propagation

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Report version: Calculation sheet with consistency tests

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1 Governing equations

For a uniform transmission line with series impedance per unit length $Z' = R + j\omega L$ and shunt admittance per unit length $Y' = G + j\omega C$,

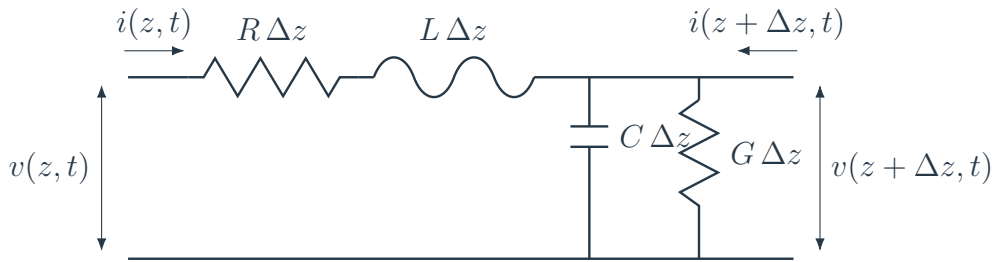
$$Z_0 = \sqrt{\frac{Z'}{Y'}}, \quad \gamma = \sqrt{Z'Y'} = \alpha + j\beta, \quad \lambda = \frac{2\pi}{\beta}. \quad (1.1)$$

For a lossless matched line, $\Gamma_L = 0$, so

$$\tilde{V}(z) = V_0^+ e^{-j\beta z}, \quad \tilde{I}(z) = \frac{V_0^+}{Z_0} e^{-j\beta z}. \quad (1.2)$$

The time convention is $e^{j\omega t}$, and all sinusoidal amplitudes below are peak values.

2 Problem 1: Lossy-line propagation



2.1 Step 1: Angular frequency and distributed quantities

$$\begin{aligned} \omega &= 2\pi(2 \times 10^9) = 1.256637061 \times 10^{10} \text{ rad/s}, \\ Z' &= 1.5 + j(1.256637061 \times 10^{10})(10 \times 10^{-9}) \\ &= 1.5 + j125.663706 \text{ } \Omega/\text{m}, \\ Y' &= 0.4 \times 10^{-3} + j(1.256637061 \times 10^{10})(0.2 \times 10^{-12}) \\ &= 0.0004 + j0.002513274 \text{ S/m}. \end{aligned}$$

2.2 Step 2: Characteristic impedance and propagation constant

$$Z_0 = \sqrt{Z'/Y'} = 221.6297 + j16.1961 \text{ } \Omega = 222.2207 \angle 4.1796^\circ \text{ } \Omega, \quad (2.1)$$

$$\gamma = \sqrt{Z'Y'} = 0.0479467 + j0.5634947 \text{ m}^{-1}. \quad (2.2)$$

2.3 Step 3: Requested secondary quantities

$$\begin{aligned}\alpha &= 0.0479467 \text{ Np/m}, \\ \alpha &= (8.6858896)(0.0479467) = 0.416465 \text{ dB/m}, \\ \beta &= 0.563495 \text{ rad/m}, \\ \lambda &= 2\pi/0.5634947 = 11.1504 \text{ m}.\end{aligned}$$

Answer. $Z_0 = (221.63 + j16.20) \Omega$; $\gamma = (0.04795 + j0.56349) \text{ m}^{-1}$;
 $\alpha = 0.4165 \text{ dB/m}$; $\lambda = 11.150 \text{ m}$.

3 Problem 2: Coaxial cable

3.1 Exact evaluation

At 500 MHz,

$$\omega = 3.141592654 \times 10^9 \text{ rad/s}, \quad Z' = 4 + j785.398163, \quad Y' = 0.001 + j0.314159265.$$

Therefore

$$Z_0 = 50.0001748 - j0.0477458 \Omega = 50.0001976 \angle (-0.0547125^\circ) \Omega, \quad (3.1)$$

$$\gamma = 0.06499997 + j15.70797043 \text{ m}^{-1}. \quad (3.2)$$

The exact attenuation in decibels is

$$\alpha_{\text{dB}} = 8.6858896(0.06499997) = 0.5645897 \text{ dB/m}.$$

3.2 Low-loss calculation

The conditions are satisfied because

$$R = 4 \ll \omega L = 785.398, \quad G = 0.001 \ll \omega C = 0.314159.$$

Consequently,

$$\begin{aligned}Z_0 &\simeq \sqrt{L/C} = \sqrt{250 \times 10^{-9} / (100 \times 10^{-12})} = 50 \Omega, \\ \alpha &\simeq \frac{1}{2} \left(\frac{4}{50} + (0.001)(50) \right) = 0.065 \text{ Np/m}, \\ \beta &\simeq (3.141592654 \times 10^9) \sqrt{(250 \times 10^{-9})(100 \times 10^{-12})} \\ &= 15.70796327 \text{ rad/m}.\end{aligned}$$

Result	Exact	Low-loss	Difference
$ Z_0 $	50.0001976 Ω	50.0000000 Ω	0.0001976 Ω
α	0.06499997	0.06500000	3.0×10^{-8}
β	15.70797043	15.70796327	7.16×10^{-6}

3.3 Phase velocity and wavelength

From the exact β ,

$$v_p = \omega/\beta = 1.99999909 \times 10^8 \text{ m/s}, \quad (3.3)$$

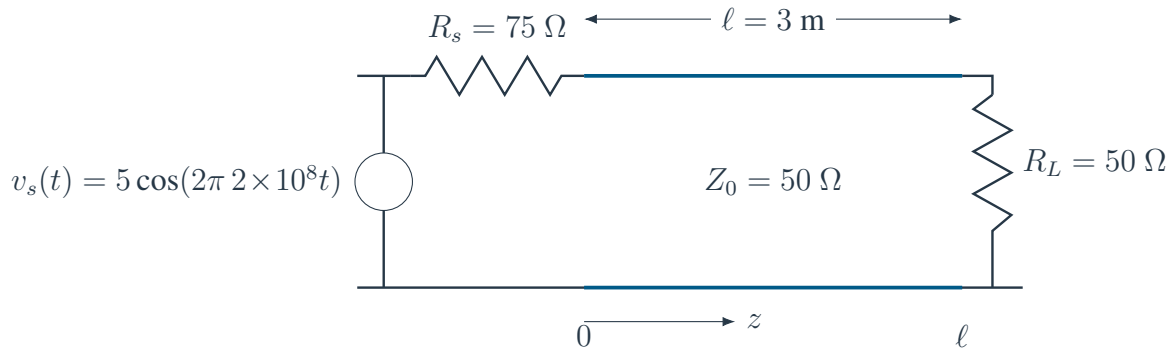
$$\lambda = 2\pi/\beta = 0.399999818 \text{ m}. \quad (3.4)$$

Rounded to engineering precision, $v_p = 2.00 \times 10^8 \text{ m/s}$ and $\lambda = 0.400 \text{ m}$.

Check: The independent low-loss relation $v_p = 1/\sqrt{LC}$ also gives $2.00 \times 10^8 \text{ m/s}$.

Answer. The cable behaves as a 50Ω low-loss line at 500 MHz. The exact γ is $(0.06499997 + j15.70797043) \text{ m}^{-1}$, and the wavelength is 0.400 m .

4 Problem 3: Three-wavelength matched line



4.1 Wave amplitude at the line input

Matching gives $Z_{\text{in}} = Z_0 = 50 \Omega$. The source divider is

$$V_0^+ = V_s \frac{Z_0}{R_s + Z_0} = 5 \frac{50}{75 + 50} = 2 \text{ V}, \quad I_0^+ = \frac{2}{50} = 0.04 \text{ A}.$$

4.2 Space-time solution

Since

$$\lambda = \frac{2 \times 10^8}{2 \times 10^8} = 1 \text{ m}, \quad \beta = 2\pi \text{ rad/m}, \quad \omega = 4\pi \times 10^8 \text{ rad/s},$$

the required instantaneous quantities are

$$v(z, t) = 2 \cos(4\pi \times 10^8 t - 2\pi z) \text{ V}, \quad (4.1)$$

$$i(z, t) = 0.04 \cos(4\pi \times 10^8 t - 2\pi z) \text{ A}. \quad (4.2)$$

The voltage-current ratio is $2/0.04 = 50 \Omega$, as it must be for a pure forward wave.

4.3 Line length and phase at the load

$$N_\lambda = \frac{3}{1} = 3, \quad \beta\ell = (2\pi)(3) = 6\pi \text{ rad.}$$

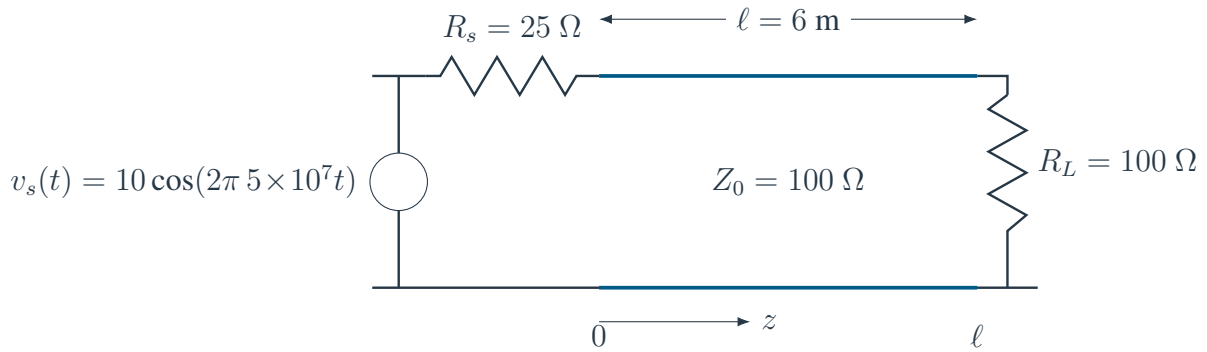
At $z = 3 \text{ m}$,

$$v_L(t) = 2 \cos(4\pi \times 10^8 t - 6\pi) = 2 \cos(4\pi \times 10^8 t) \text{ V.}$$

Thus the load and source sinusoids are in phase; the load peak amplitude is $2/5 = 0.4$ times the ideal source amplitude because of the source resistance.

Answer. The line length is 3λ (1080°). The load voltage is 2 V peak at the same phase as the source.

5 Problem 4: One-wavelength air line



5.1 Matched termination

The frequency is 50 MHz. In air,

$$\lambda = \frac{3 \times 10^8}{5 \times 10^7} = 6 \text{ m}, \quad \beta = \frac{2\pi}{6} = \frac{\pi}{3} \text{ rad/m}, \quad \omega = \pi \times 10^8 \text{ rad/s.}$$

Since the load is matched,

$$V_0^+ = 10 \frac{100}{25 + 100} = 8 \text{ V}, \quad I_0^+ = 8/100 = 0.08 \text{ A.}$$

Hence, for $0 \leq z \leq 6 \text{ m}$,

$$v(z, t) = 8 \cos\left(\pi \times 10^8 t - \frac{\pi}{3} z\right) \text{ V}, \quad (5.1)$$

$$i(z, t) = 0.08 \cos\left(\pi \times 10^8 t - \frac{\pi}{3} z\right) \text{ A.} \quad (5.2)$$

The average power is constant everywhere on a lossless matched line:

$$P_{\text{av}} = \frac{1}{2} V_0^+ I_0^+ = \frac{1}{2} (8)(0.08) = 0.320 \text{ W.} \quad (5.3)$$

5.2 Open-circuit termination

The matched expression cannot be retained. For an open circuit, $\Gamma_L = +1$, so $|V^-| = |V^+|$. Since the line is exactly one wavelength, the open circuit is repeated at the input: $Z_{\text{in}} \rightarrow \infty$. No current flows through R_s , and the input voltage has the full 10 V peak value. Therefore $V_0^+ = V_0^- = 5$ V at $z = 0$, and

$$v(z, t) = \Re \{ 5 (e^{-j\beta z} + e^{j\beta z}) e^{j\omega t} \} \quad (5.4)$$

$$= 10 \cos\left(\frac{\pi z}{3}\right) \cos(\pi \times 10^8 t) \text{ V}. \quad (5.5)$$

This result has fixed voltage nodes and antinodes, which identifies it as a standing wave. At the load $z = 6$ m, the current is zero and $v_L(t) = 10 \cos(\pi \times 10^8 t)$ V.

Answer. Matched case: $P_L = 0.320$ W. Open-circuit case: the single-wave equation is invalid and must be replaced by the superposition of incident and reflected waves.

6 Answer checklist

Problem	Verified result
1	$\alpha = 0.0479467$ Np/m = 0.416465 dB/m, $\beta = 0.563495$ rad/m, $\lambda = 11.1504$ m.
2	Exact and low-loss values agree to at least four significant figures; $v_p \simeq 2.00 \times 10^8$ m/s, $\lambda \simeq 0.400$ m.
3	$\ell = 3\lambda$, $V_L = 2\angle 0^\circ$ V (peak phasor).
4	$\ell = \lambda$, matched $P_L = 0.320$ W; open circuit gives a standing wave.